

^{60}Ga $\varepsilon+\beta^+$ decay: XUNDL-5 2025Ow01

Parent: ^{60}Ga : $E=0$; $J^\pi=2^+$; $T_{1/2}=69.4$ ms 2; $Q(\varepsilon+\beta^+)=14160$ 15; $\% \varepsilon+\beta^+$ decay=100.0

$^{60}\text{Ga}-J^\pi$: from the Adopted Levels of ^{60}Ga .

$^{60}\text{Ga}-T_{1/2}$: unweighted average of 70 ms 15 (2001Ma96), 70 ms 13 (2002Lo13), 76 ms 3 (2017Ku12), 70.8 ms 20

(2017GoZT,2020Gi02), 69.4 ms 2 (2021Or01), and 57.7 ms 98 (2025Ow01).

$^{60}\text{Ga}-Q(\varepsilon+\beta^+)$: deduced by evaluators using ^{60}Zn mass excess of -54174.5 5 (2021Wa16) and ^{60}Ga mass excess of -40015 15; weighted average of -40016 15 (2021Or01), -40005 30 (2021Pa44), and -40034 46 (2023Wa10).

Compiled (unevaluated) dataset from 2025Ow01: Phys Rev C 111, 065801 (2025).

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2025Ow01: ^{60}Ga was produced via the projectile fragmentation of a 150-MeV/nucleon $^{78}\text{Kr}^{36+}$ primary beam impinging on a 517-mg/cm² thick ^9Be target, filtered by the NSCL A1900 separator and the radio frequency fragment separator, identified using the ΔE -ToF method, and implanted into a 1-mm-thick DSSD with 16×16 1.2-mm wide strips at the center of the Summing NaI (SuN) detector. Beam ions and decay positrons were detected using the DSSD and γ rays were detected using the SuN detector. Measured implant-decay correlations, γ -ray total absorption spectra (TAS), sum of segments spectra (SoS), and multiplicity spectra. Deduced β -feeding intensities, Gamow-Teller strength distributions, and $T_{1/2}$ of parent ^{60}Ga . Compared with shell-model calculations using GXPF1A and KB3G Hamiltonians.

 ^{60}Zn Levels

<u>E(level)[†]</u>	<u>J^π</u>	<u>E(level)[†]</u>	<u>E(level)[†]</u>	<u>E(level)[†]</u>
0	0 ⁺	3808	4913	≈7683
1004		3812	5200	≈8067
2193		3972	≈5460	≈8470
2559		4180	≈5733	≈8894
3035		4200	≈6020	≈9338
3200		4351	≈6321	≈9805
3510		4400	≈6637	
3627		4776	≈6969	
3710		4852	≈7317	

[†] For levels below 5.3 MeV, 2025Ow01 adopts E(level) from the RIPL-3 library, originally from the Adopted Levels of ^{60}Zn (2013Br15) rounded to the nearest keV. For levels above 5.3 MeV, E(level) is a sample of calculated continuum-state energies with energy spacing (273 keV) that follows the energy resolution of the SuN detector.

 ε, β^+ radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>$I_{\beta^+}^{\dagger @}$</u>	<u>$I_{\varepsilon}^{\dagger @}$</u>	<u>$\text{Log } ft^{\dagger}$</u>	<u>$I(\varepsilon+\beta^+)^{\dagger @}$</u>	<u>Comments</u>
(4.36×10 ³)	≈9805	0.49	0.01	3.8	0.5	$I(\varepsilon+\beta^+)$: 0.5 +13–4.
(4.82×10 ³)	≈9338	1.18	0.02	3.7	1.2	$I(\varepsilon+\beta^+)$: 1.2 +9–12.
(5.27×10 ³)	≈8894	0.10	9.077×10 ^{−4}	5.0	0.1	$I(\varepsilon+\beta^+)$: 0.1 +4–1.
(5.69×10 ³)	≈8470	1.69	0.01	3.9	1.7	$I(\varepsilon+\beta^+)$: 1.7 +13–11.
(6.09×10 ³)	≈8067				<0.1 [#]	
(6.48×10 ³)	≈7683				<0.1 [#]	$I(\varepsilon+\beta^+)$: 0.0 +3–0.
(6.84×10 ³)	≈7317				<0.1 [#]	
(7.19×10 ³)	≈6969				<0.1 [#]	
(7.52×10 ³)	≈6637	0.10	2.641×10 ^{−4}	5.8	0.1	$I(\varepsilon+\beta^+)$: 0.1 +9–1.
(7.84×10 ³)	≈6321				<0.1 [#]	
(8.14×10 ³)	≈6020	1.00	0.002	5.0	1.0	$I(\varepsilon+\beta^+)$: 1.0 +74–10.
(8.43×10 ³)	≈5733	4.89	0.01	4.4	4.9	$I(\varepsilon+\beta^+)$: 4.9 +13–49.
(8.70×10 ³)	≈5460	3.09	0.005	4.7	3.1	$I(\varepsilon+\beta^+)$: 3.1 +21–8.
(8960 15)	5200	4.69	0.01	4.5	4.7	$I(\varepsilon+\beta^+)$: 4.7 +15–47.
(9247 15)	4913	11.0	0.01	4.2	11.0	$I(\varepsilon+\beta^+)$: 11.0 +22–41.
(9308 15)	4852	37.7	0.05	3.7	37.7	$I(\varepsilon+\beta^+)$: 37.7 +75–47.
(9384 15)	4776				<0.1 [#]	

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^{60}Ga $\varepsilon+\beta^+$ decay: XUNDL-5 2025Ow01 (continued) ε, β^+ radiations (continued)

E(decay)	E(level)	$I\beta^+ \text{ }^\dagger @$	$I\varepsilon \text{ }^\dagger @$	$\text{Log } ft \text{ }^\dagger$	$I(\varepsilon + \beta^+) \text{ }^\ddagger @$	Comments
(9760 15)	4400				<0.1 [#]	
(9809 15)	4351				<0.1 [#]	
(9960 15)	4200				<0.1 [#]	
(9980 15)	4180				<0.1 [#]	
(10188 15)	3972	8.59	0.01	4.6	8.6	$I(\varepsilon + \beta^+)$: 8.6 +28-16.
(10348 15)	3812	5.30	0.005	4.8	5.3	$I(\varepsilon + \beta^+)$: 5.3 +14-52.
(10352 15)	3808				<0.1 [#]	
(10450 15)	3710				<0.1 [#]	
(10533 15)	3627	5.79	0.005	4.8	5.8	$I(\varepsilon + \beta^+)$: 5.8 +11-12.
(10650 15)	3510	2.10	0.002	5.3	2.1	$I(\varepsilon + \beta^+)$: 2.1 +6-7.
(10960 15)	3200				<0.1 [#]	
(11125 15)	3035	0.40	3.017×10^{-4}	6.1	0.4	$I(\varepsilon + \beta^+)$: 0.4 +3-4.
(11601 15)	2559	7.79	0.005	4.9	7.8	$I(\varepsilon + \beta^+)$: 7.8 +10-12.
(11967 15)	2193				<0.1 [#]	
(13156 15)	1004	0.2		6.8	0.2	$I(\varepsilon + \beta^+)$: 0.2 +13-2.
(14160 15)	0	3.8 6	0.0027 4	8.3 1	3.8 6	

[†] Calculated by evaluators using the BetaShape code (2023Mo21).

[‡] 2025Ow01 extracts $I(\varepsilon + \beta^+)$ at each E(level) by simultaneously fitting the simulated TAS, SoS, and multiplicity spectra to data via the χ^2 minimization. The sum is normalized to $\Sigma I(\varepsilon + \beta^+) = 100$.

[#] $I(\varepsilon + \beta^+)$ values below 0.1 were set to 0.0 in 2025Ow01 due to sensitivity limitations.

[@] Absolute intensity per 100 decays.